Mahdi Kallel

PhD candidate in reinforcement learning

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Profile

Graduating from the University of Würzburg in April 2027, advised by Carlo D'Eramo. I started out designing algorithms for exploration and policy optimization, and now study the learning dynamics of neural networks, in particular memory and plasticity in continual learning, and how reinforcement learning changes them compared with supervised learning.

Publications

2026 **Mahdi Kallel**, Nour Jamoussi, Johannes Tölle, Marios Kountouris, Carlo D'Eramo. **The Spectral Anatomy of Plasticity Loss**. *AXIOM Workshop at NeurIPS 2026*.

- Uses random matrix theory to explain loss of plasticity.

→ Over a stream of tasks, the directions that hold earlier knowledge absorb a growing share of each new gradient. Making them read-only keeps the network trainable without losing what it learned.

2026 **Mahdi Kallel**, Johannes Tölle, Ahmed Hendawy, Carlo D'Eramo. **Do Not Imitate, Reinforce: Iterative Classification via Belief Refinement**. *Transactions on Machine Learning Research (TMLR), 2026*.

- Trains classifiers with reinforcement learning instead of imitating labels with cross-entropy: a recurrent agent refines its prediction step by step and is rewarded for each improvement.

→ It achieves a better balance of accuracy and calibration than supervised training, and stays calibrated under human label noise, where cross-entropy grows overconfident.

2026 **Mahdi Kallel**, Jose-Luis Holgado-Alvarez, Samuele Tosatto, Carlo D'Eramo. **PPO+: Enhancing Proximal Policy Optimization**. *Reinforcement Learning Journal (RLC 2026)*.

- Shows that PPO is held back by its critic.

→ Training the critic on off-policy data, while keeping the policy update on-policy, substantially improves PPO and narrows the gap to off-policy algorithms, without implementation tricks.

2026 Ahmed Hendawy, Henrik Metternich, Théo Vincent, **Mahdi Kallel**, Jan Peters, Carlo D'Eramo. **Use the Online Network If You Can: Towards Fast and Stable Reinforcement Learning**. *ICLR 2026*.

2024 **Mahdi Kallel**, Debabrota Basu, Riad Akrour, Carlo D'Eramo. **Augmented Bayesian Policy Search**. *ICLR 2024*.

- Makes Bayesian policy search aware that it is solving an MDP: the performance difference lemma turns a learned critic into the prior over returns.

→ The method outperforms earlier Bayesian optimization approaches, with the largest gains on high-dimensional policies.

2024 Oliver Järnefelt, **Mahdi Kallel**, Carlo D'Eramo. **Cyclicity-Regularized Coordination Graphs**. *RLC 2024*.

Research Experience

2022 **Research Intern**, *INRIA Lille, SCOOL Research Group*, France
- Started the work on Bayesian policy search with deterministic policies that became Augmented Bayesian Policy Search (ICLR 2024).

2020 **Research Intern**, *EURECOM*, Biot, France
- Compared centralized and decentralized approaches to mobile traffic prediction and load balancing.

Industry Experience

2021 **Machine Learning Engineer**, *Mcovision (independent)*, Paris, France
- Built a classifier of echocardiography videos for cardiac amyloidosis and other rare heart diseases with Hôpital Cochin, from data cleaning to training.

2021 **Research Intern**, *SustainEcho*, Paris, France
- Explored few-shot and hierarchical learning methods for classification with little data.

Education

2023–2027 **PhD, Reinforcement Learning**, *University of Würzburg*, Germany
Expected April 2027. Supervisor: Prof. Carlo D'Eramo.
Teaching: Introduction to Machine Learning; Introduction to Reinforcement Learning.

2021–2022 **MSc, Optimization**, *Université Paris-Saclay*, France
Coursework: Convex analysis, dynamic programming, stochastic optimization and approximation.

2020–2021 **MSc, Mathematics, Vision, Learning (MVA)**, *ENS Paris-Saclay*, France
Coursework: Reinforcement learning, convex optimization, kernel methods, advanced deep learning.

2018–2020 **BSc, Data Science**, *Télécom Paris / EURECOM*, France

Supervision, Talks and Service

Supervision Johannes Tölle, MSc thesis, University of Würzburg (2026). The thesis received the VDE Bayern Award 2026, and Johannes co-authored two of the papers above.

Talks *On Better Priors for Policy Search*. Invited talk and lab visit, Improbable AI Lab (Pulkit Agrawal), MIT, August 2024.

Reviewer NeurIPS (top reviewer, 2026), ICLR, ICML, RLC

Technical Skills

Programming Python, JAX, PyTorch, NumPy, Julia, C

Languages Arabic (native), English (fluent), French (fluent), German (basic)